1 Given that $P(A) = 0.9$, find $P(A')$

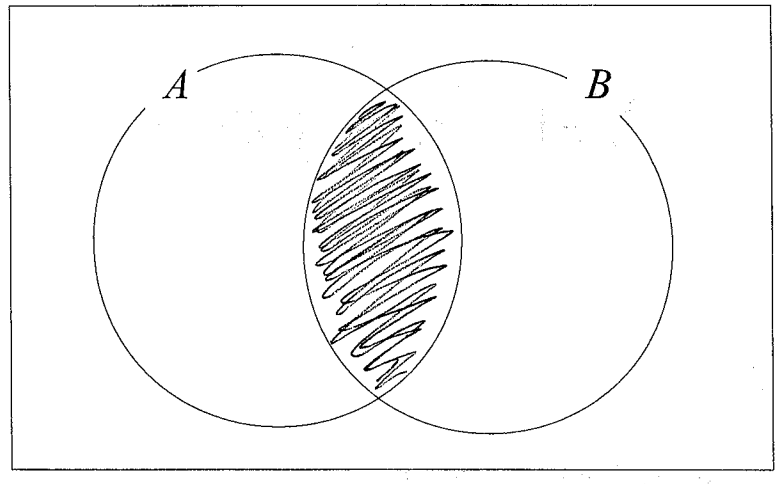
$1 - 0.9$

0.1

(Total for question 1 is 1 mark)

---

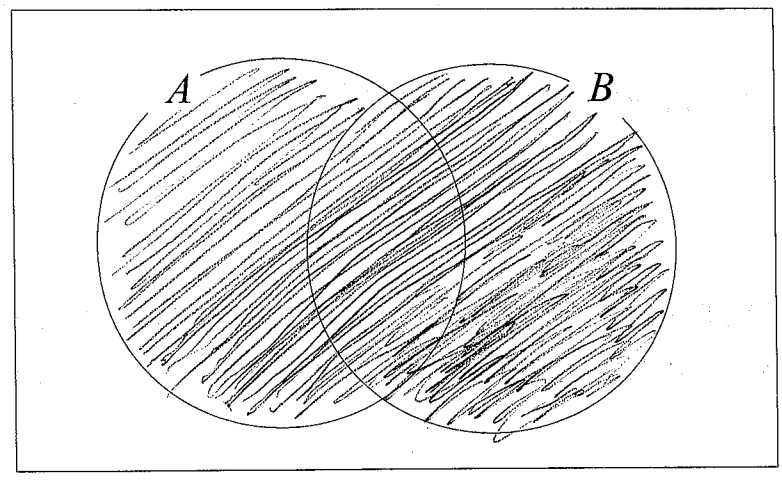
2 Shade the region that represents $(A \cap B)$



(Total for question 2 is 1 mark)

---

3 Shade the region that represents $(A \cup B)$



(Total for question 3 is 1 mark)

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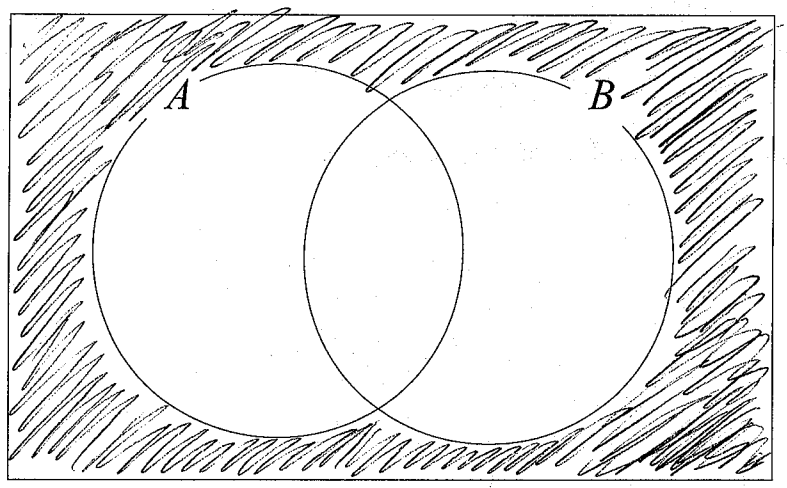
4 Shade the region that represents $(A' \cap B')$



(Total for question 4 is 1 mark)

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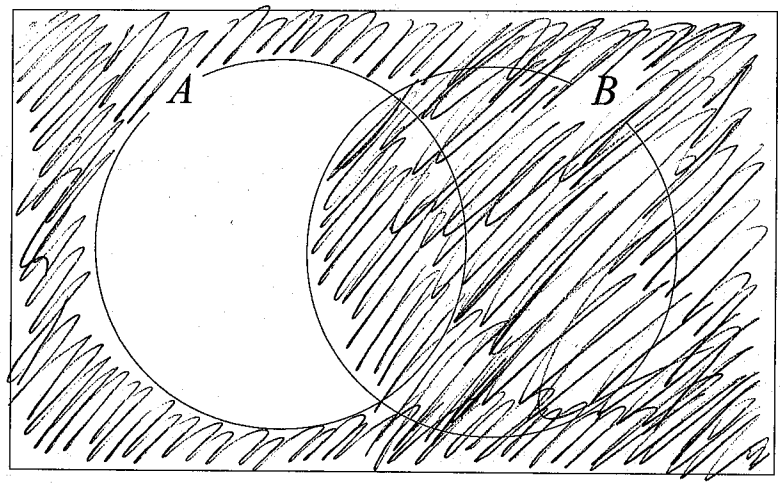
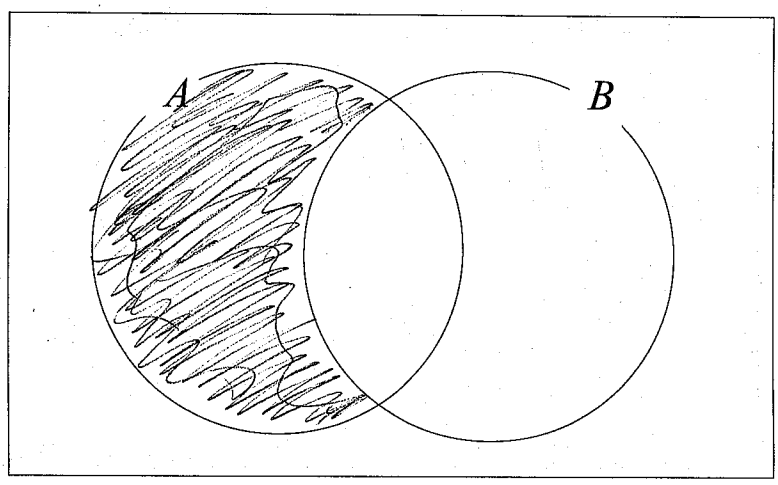
5 Shade the region that represents $(A' \cup B)$



(Total for question 5 is 1 mark)

---

6 Shade the region that represents $(A \cap B')$

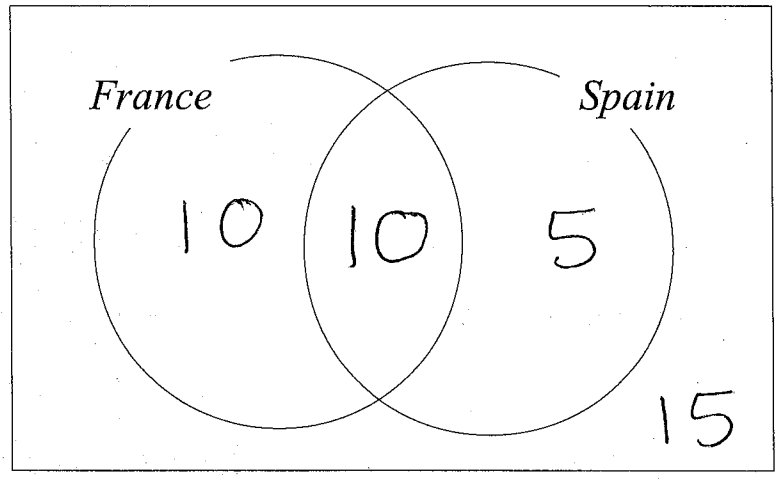


(Total for question 6 is 1 mark)

7 40 students were surveyed:

20 have visited France
15 have visited Spain
10 have visited both France and Spain

Use this information to complete the Venn Diagram



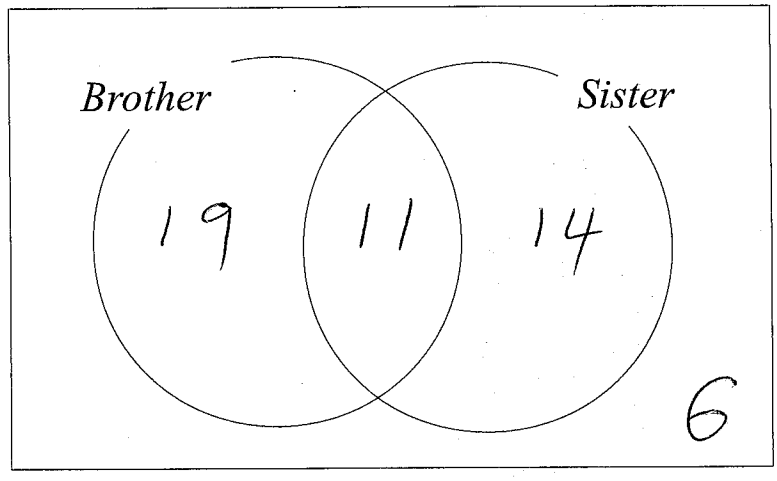
(Total for question 7 is 3 marks)

8 Out of 50 people surveyed:

30 have a brother 55
25 have a sister
6 have neither a brother or a sister

Use this information to complete the Venn Diagram

55 - 44 = 11



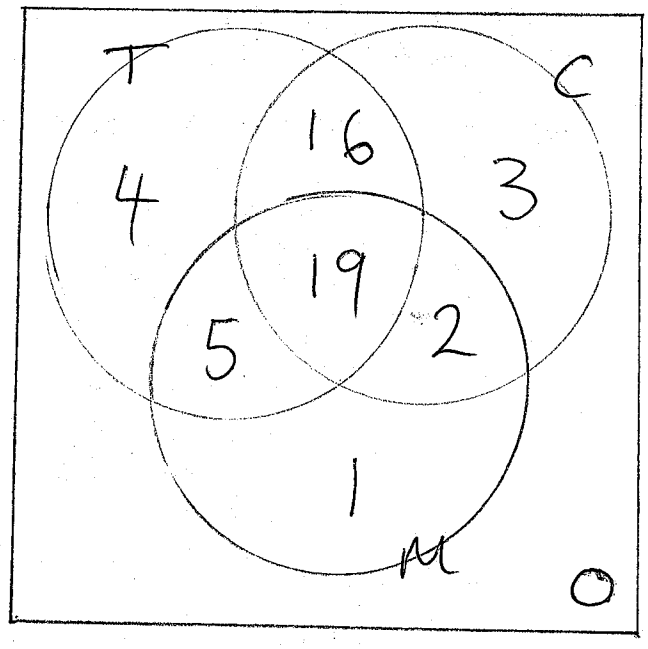
(Total for question 8 is 3 marks)

9 Sami asked 50 people which drinks they liked from tea, coffee and milk.

All 50 people like at least one of the drinks
19 people like all three drinks.
16 people like tea and coffee but do not like milk.
21 people like coffee and milk.
24 people like tea and milk.
40 people like coffee.
1 person likes only milk.

Sami selects at random one of the 50 people.

Work out the probability that this person likes tea.



$\frac{44}{50}$

.....

(Total for question 9 is 4 marks)

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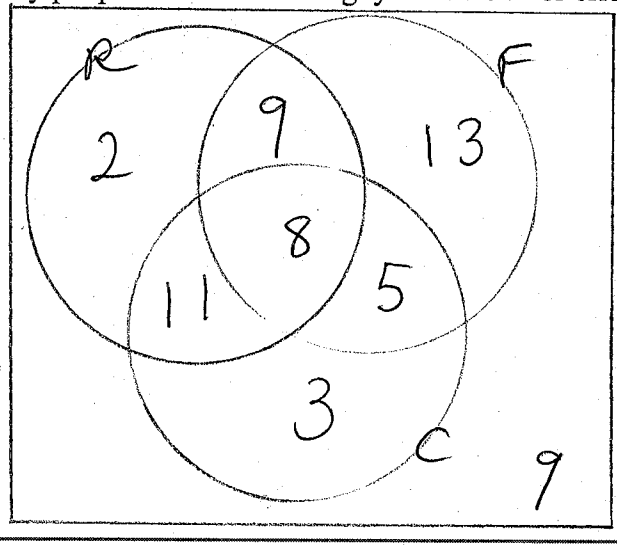
10 Sami asked 60 people which sports they liked from rugby, football and cricket.

8 people like all three sports.
17 people like rugby and football.
13 people like football and cricket.
19 people like rugby and cricket.
35 people like football.
27 people like cricket
30 people like rugby.

a) How many people liked neither rugby or football or cricket?



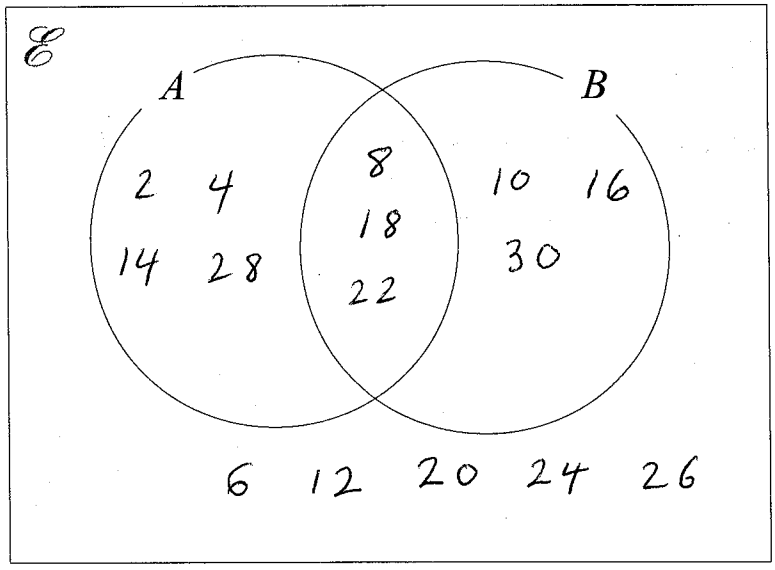
9

.....

(Total for question 10 is 4 marks)

11 $\mathscr{E}$ = {even numbers between 1 and 31}
A = {2, 4, 8, 14, 18, 22, 28}
B = {8, 10, 16, 18, 22, 30}

(a) Complete the Venn diagram to represent this information.



(4)

A number is chosen at random from the universal set, $\mathscr{E}$

(b) What is the probability that the number is in the set $A \cup B$?

$\frac{10}{15}$

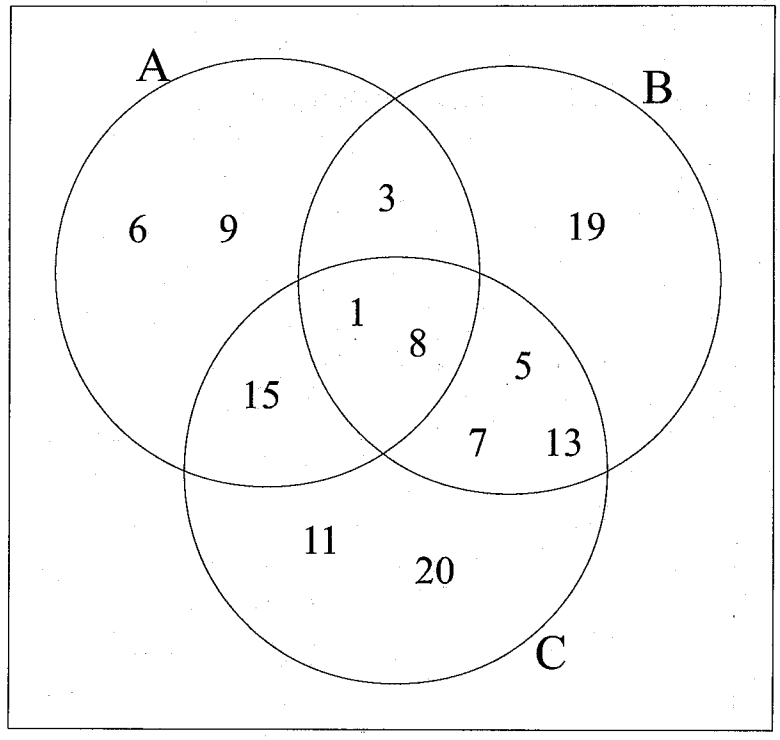
.....

(2)

(Total for question 11 is 6 marks)

$\left[\frac{2}{3}\right]$

12 Here is a Venn diagram.



(a) List the members of A ∩ B

1, 3 and 8

(1)

A number is chosen at random from 8.

(b) Find $P(B \cup C)$

$\frac{10}{12}$

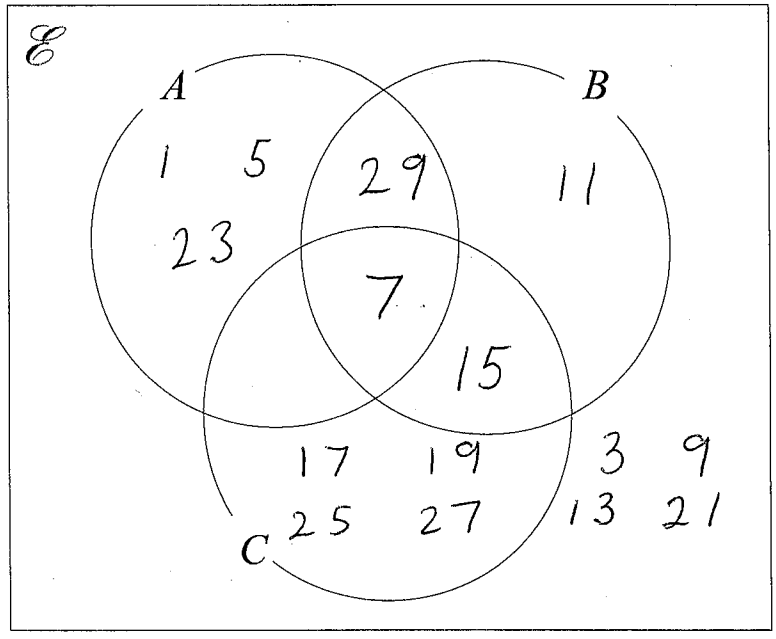
(2)

(Total for question 12 is 3 marks)

$\left[\frac{5}{6}\right]$

13 $\mathscr{E}$ = {odd numbers less than 30}
A = {1, 5, 7, 23, 29}
B = {7, 11, 15, 29}
C = {7, 15, 17, 19, 25, 27}

(a) Complete the Venn diagram to represent this information.



A number is chosen at random from $\mathscr{E}$.

(b) Find the probability that the number is a member of $(A \cap B)$.

$\frac{2}{15}$

(Total for question 26 is 3 marks)